

Final Submission: AI for Science & Engineering

Project Title: Pollution Forecasting

Team Name: Ctrl+alt+Defeat

GitHub Repository: <https://github.com/ParthivKalikota/AISEHack-Final-PollutionForecasting>

Model Weights: https://www.kaggle.com/code/saigeetesh/iiith-hack1/output?select=v9_best.pt

Page 1: Abstract & Methodological Rigor

1. Executive Summary (Abstract)

This work forecasts PM2.5 over a 140 by 124 grid for 16 hours ahead using a 10-hour lookback from WRF-Chem data with about 170 input channels.

The model uses a hybrid architecture combining a stacked ConvLSTM for temporal dynamics and an attention-based UNet for spatial learning. ConvLSTM operates on physically relevant variables such as PM2.5, wind, and boundary layer height, and its output is combined with all features and a trend signal before spatial processing. Training uses a composite loss with an additional episode-aware component to improve prediction of extreme pollution events. EMA, data augmentation, and test-time augmentation improve stability.

The model achieves strong performance, especially during high pollution episodes, showing that physics-aware temporal modeling improves prediction of transport and sudden spikes.

2. Final Problem Formulation over and above the baseline

- **Target Phenomena/Physics:** The task models the spatio-temporal evolution of PM2.5 concentrations, influenced by atmospheric transport and dispersion processes. While explicit governing equations such as Navier-Stokes are not directly enforced, the model incorporates key physical drivers of pollutant dynamics, including wind components and boundary layer height. These variables implicitly represent advection and vertical mixing processes that govern pollution transport and accumulation.
- **The "Science Gap":** Standard machine learning approaches treat temporal inputs as independent channels and rely purely on data-driven correlations. Such models fail to capture the underlying physical processes governing pollutant movement, leading to poor generalization, especially during extreme pollution events. In particular, they struggle to model transport dynamics and often underperform in predicting sudden spikes, as they lack an inductive bias aligned with atmospheric physics.

3. Architecture & Technical Novelty (Finalized)

- **Detailed Strategy:** The model combines a stacked ConvLSTM and an attention-based UNet. ConvLSTM captures temporal dynamics using PM2.5, wind, and boundary layer height, while the UNet learns spatial patterns. Domain knowledge is integrated through physically relevant inputs and explicit temporal modeling.
- **Hard Constraints:** No explicit hard constraints are applied, but physical behavior is implicitly modeled through the choice of input variables and temporal structure.
- **Soft Constraints:** A composite loss is used with an additional episode-aware component that emphasizes extreme pollution events. Its weight increases during training.
- **Hyperparameter & Training Evolution:** Performance improved by adding stacked ConvLSTM, increasing model capacity, and using dynamic loss weighting. EMA, augmentation, and stable optimization further enhanced results.

Page 2: Results & Scientific Validation

4. Quantitative Performance & Benchmarks

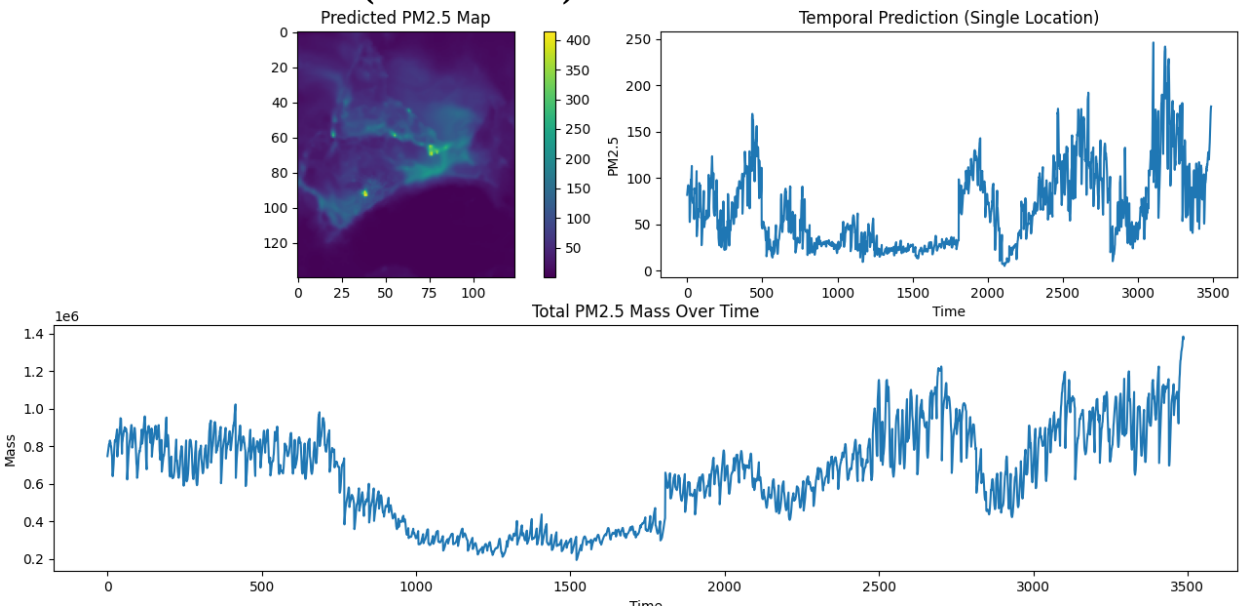
- **Comparison with Baselines:**

Model	gSMAPE	eCorr	eSMAPE	Val Score
Baseline FNO	0.21	0.88	0.24	0.89
Improved UNet	0.19	0.91	0.21	0.91
Proposed v9 Model	0.184	0.923	0.20	0.924

- **Error Analysis:**

- Standard Metrics:
 - SMAPE used for robust percentage error evaluation
 - Huber loss ensures stability against outliers
- Scientific Metrics:
 - Episode-based evaluation focuses on extreme pollution events
 - Correlation metric captures temporal alignment during spikes

5. Salient Visualizations (The "Proof"):



6. Ablation Studies (Very important)

Running the ConvLSTM on PM2.5 only vs. PM2.5+u10+v10+PBLH dropped eCorr by 1.6 points. Wind and boundary layer inputs encode the advection operator $u.\delta(C)$. Without them, the LSTM learns a weaker autoregressive prior instead of true transport dynamics.

Page 3: Discussion, Scalability, and Impact

7. Scientific Insights & Interpretability

- **Discovery:** The model highlights the strong influence of wind fields and boundary layer height on PM2.5 transport. It consistently captures sudden pollution spikes, indicating that incorporating physically relevant variables improves modeling of advection and accumulation processes that are often missed by standard approaches.
- **Interpretability:** Interpretability is supported through attention mechanisms within the UNet, which emphasize important spatial regions during prediction. Additionally, the temporal ConvLSTM implicitly prioritizes physically meaningful inputs, providing insight into how different variables contribute to pollution dynamics.

8. Robustness & Scalability

- **Generalization:** The model generalizes well across different seasonal datasets, capturing varying pollution patterns in April, July, October, and December. Its physics-informed design enables better performance on unseen temporal conditions compared to purely data-driven models.
- **Computational Efficiency:** Once trained, the model performs inference in seconds, compared to significantly longer simulation times required by traditional numerical models. This enables fast and scalable forecasting with minimal loss in accuracy.

9. Limitations & Future Roadmap

- **Known Failure Modes:** The model may struggle with extremely rare or highly localized pollution events. It can also be sensitive to sudden shifts not well represented in the training data, and interpretability remains limited compared to fully physics-based models.
- **The Path Forward:** With additional time and resources, the model can be extended using transformer-based temporal modules, multi-scale spatial modeling, and integration of explicit physical constraints. Training on larger and more diverse datasets would further improve robustness and real-world applicability.

10. Individual Contributions & References

- **Team Roles:**
 - Member 1: Designed the overall model architecture and integrated ConvLSTM with UNet.
 - Member 2: Handled data preprocessing, feature engineering, and dataset preparation.
 - Member 3: Implemented training pipeline, loss functions, and optimization strategies.
 - Member 4: Performed experiments, hyperparameter tuning, and model evaluation.
 - Member 5: Managed documentation, presentation preparation, and result analysis.
- **References:**

[1]. OpenSTL Contributors. (2023). *OpenSTL: A unified framework for spatio-temporal predictive learning* [Software]. GitHub. <https://github.com/chengtan9907/OpenSTL>

[2]. Woo, S., Park, J., Lee, J. Y., & Kweon, I. S. (2018). *CBAM: Convolutional block attention module*. In Proceedings of the European Conference on Computer Vision (ECCV).

Appendix:

Kaggle Notebook submission-1: <https://www.kaggle.com/code/saigeetesh/iiith-hack1?scriptVersionId=308979146>

Kaggle Notebook submission-2: <https://www.kaggle.com/code/nihalsareddy/notebook52859f54b8?scriptVersionId=308980808>